

Graphs:

- Non-linear data structure
- Having vertices & edges
- More than one ways to move from one node to other
- Every tree is a graph but every graph is not a tree

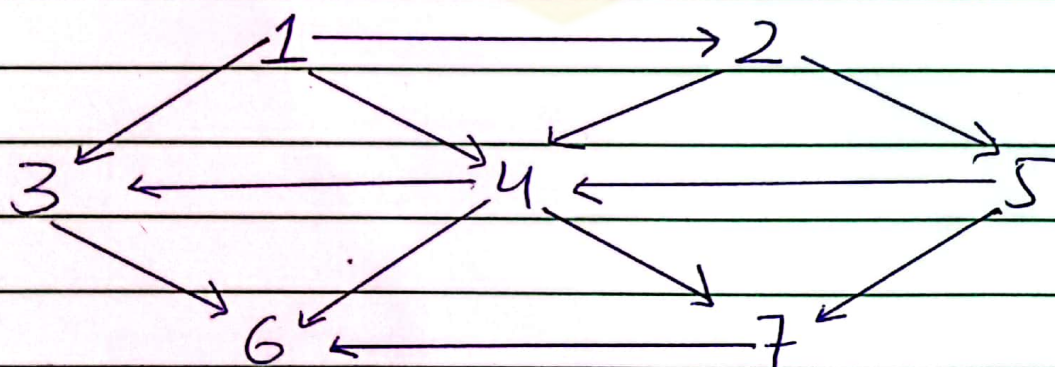
-: Traversal of Graph :-

• BFS

(Breadth First Search)

Solved by using Queue:

from vertex 2



2 | 4 | 5

We have visited 2

Result: 2

4 | 5 | 3 | 6 | 7

Result:

2, 4

5 | 3 | 6 | 7

5 is connected to 4 & 7
4 is already visited & 7 is
already in queue

Result

2, 4, 5

3 | 6 | 7

Result

2, 4, 5, 3

6 | 7

Result

2, 4, 5, 3, 6

7

Result

2, 4, 5, 3, 6, 7

Types of graph

DATE: _____

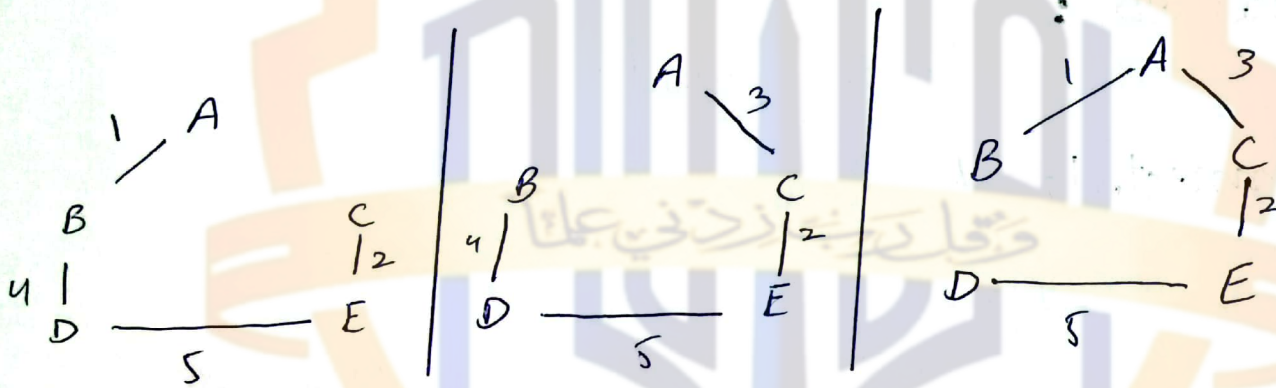
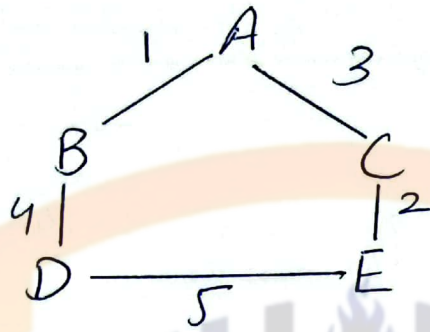
- undirected :
 - Edges have no direction
 - bidirectional.
- directed :
 - Have direction
- connected: A graph is connected if we reach from any vertex to other vertex by following edges.
 - Not compulsory to be cyclic

Spanning tree

- Subgraph of an connected graph
- It includes all vertices with least possible no. of edges
- It can not be disconnected.
- It does not have cycles.
- consist of n no. of vertices & $(n-1)$ no. of edges
- n^{n-2} no. of graph (spanning) can be

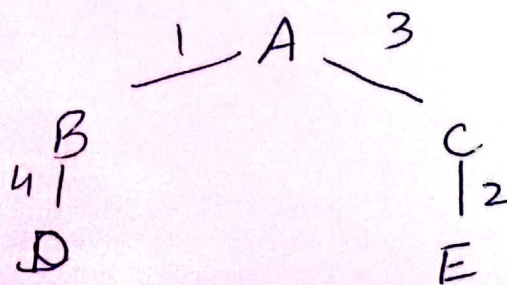
Al-Kamal Notes

created from a complete graph
Example



weight does not matter in spanning tree but weight does matter in minimum spanning tree.

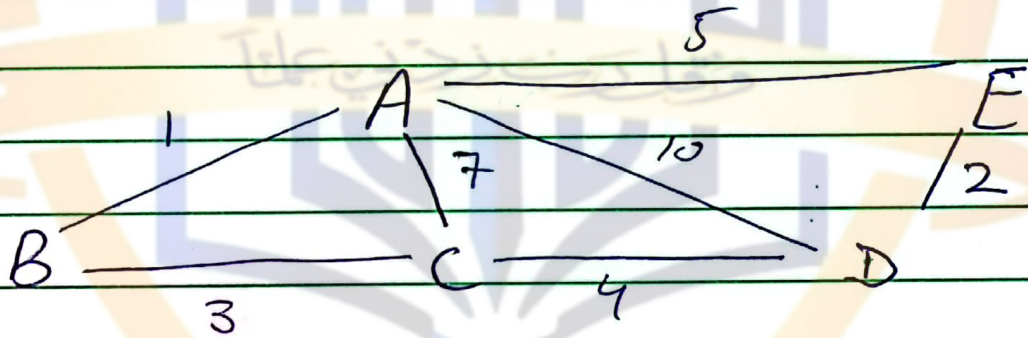
MST: In MST we will remove those edge that have highest weight from a cyclic graph



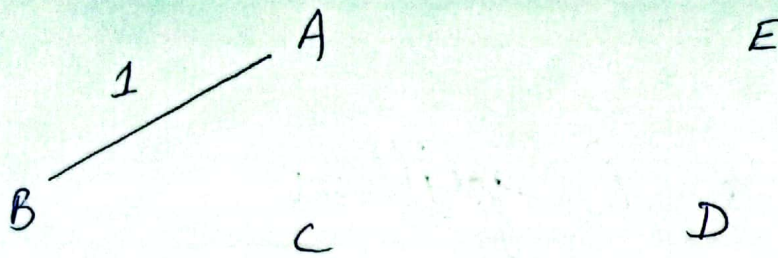
Kruskal's Algorithm

DATE: _____

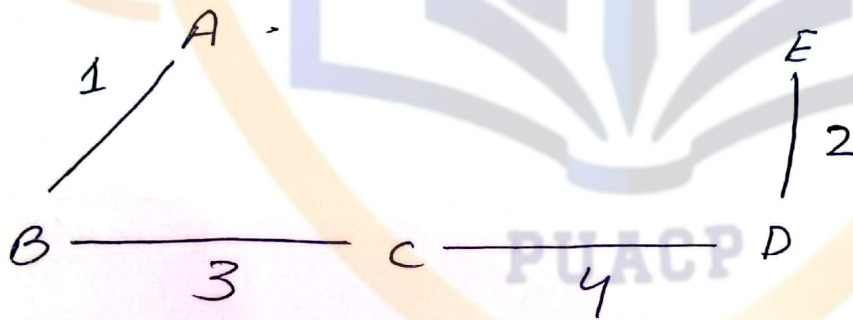
- First sort all the edges from low weight to high.
- Now take the edges with lowest weight and ^{add} it to spanning tree.
- if any edge creates cycle then reject it.



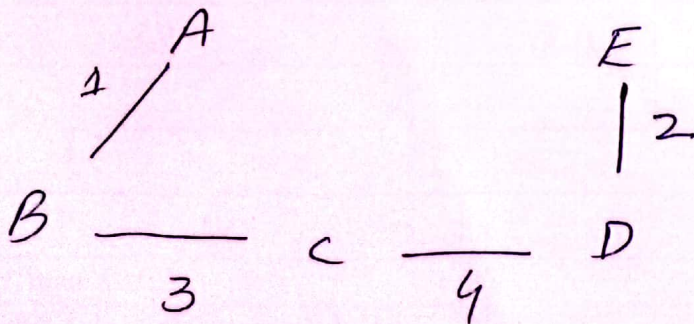
Edge	weight	Edge	weight
AB	1	AB	1
AC	7	DE	2
AD	10	BC	3
AE	5	CD	4
BC	3	AE	5
CD	4	AC	7
DE	2	AD	10



make edges with weight 1



we won't make 5 & 7, 10 because the make cyclic path



No. of vertices = 5

No. of edges = $5 - 1 = 4$

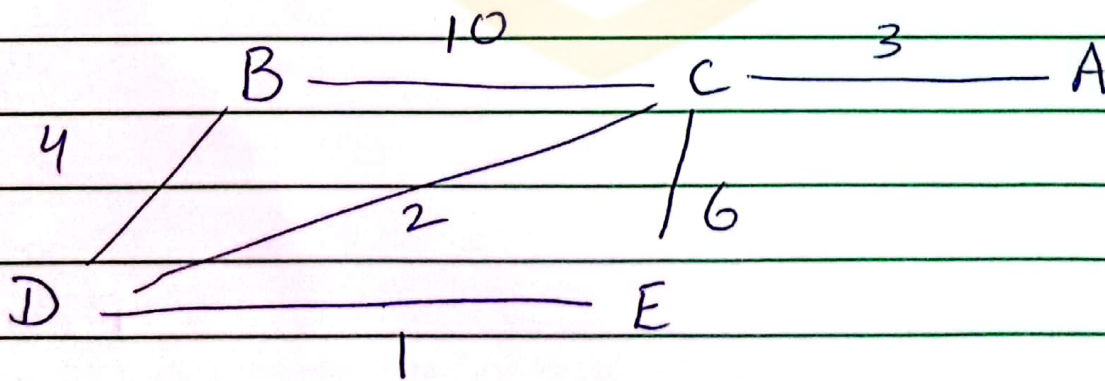
minimum count = $1 + 3 + 4 + 2 = 10$

Prim's Algorithm:

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- First we choose a vertex & check its possible edges we will make the edge with smaller cost
- Now we will move to next point by the edge we made. Now we will compare its cost all edges with remaining edges of visited vertex.
- Select the edge with lowest cost
- If any edge make graph cyclic we will not select that edge.

Example:

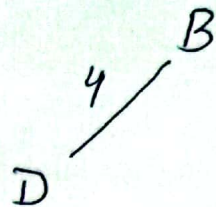


Step 1:

Select B

B-10
4 1

choose lowest



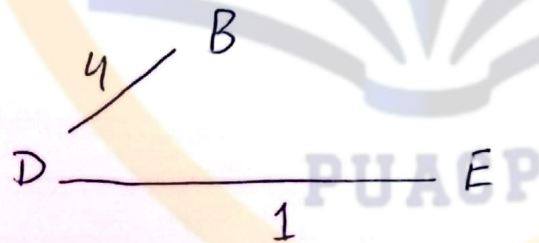
Remaining	cost
BC	10

Step 2:

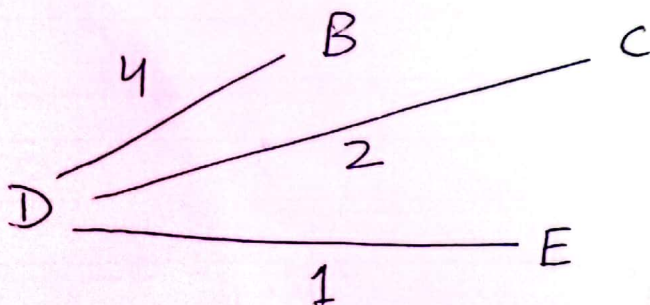


D has 2 & 1 cost

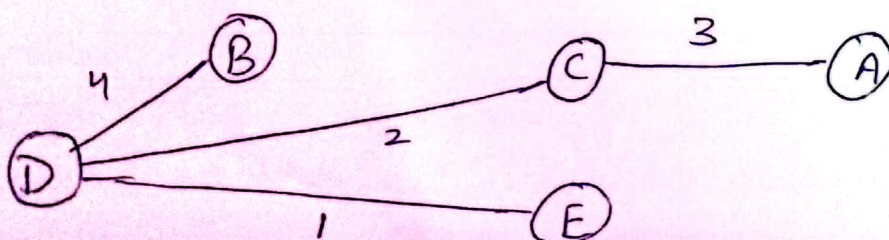
compare 2, 1 and remaining edge cost of 10



Remaining	cost
BC	10
DC	2



Remaining	cost
BC	10
EC	6



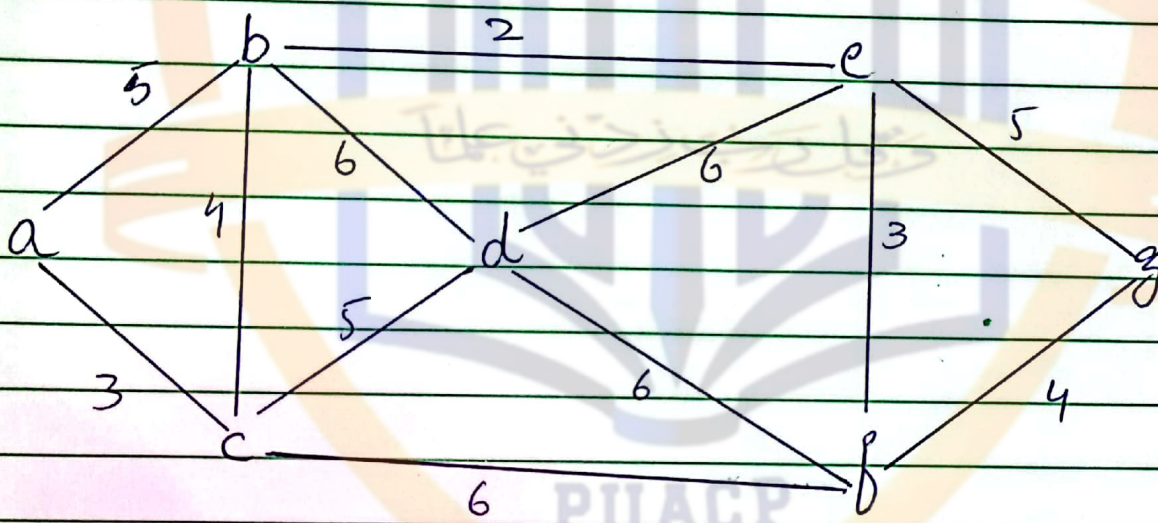
Dijkstra's Algorithm:

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• Used to find shortest path

• It is minimization problem

↓
Solved by Greedy Algorithm



Start from node 'b'

Source	Destination					
b	a	c	d	e	f	g
b	5	4	6	(2)	∞	∞
b, e	5	(4)	6	(2)	5	7
b, e, c	(5)	(4)	6	(2)	5	7
b, e, c, a	(5)	(4)	6	(2)	(5)	7
b, e, c, a, f	(5)	(4)	(6)	(2)	(5)	7
b, e, c, a, f, d	(5)	(4)	(6)	(2)	(5)	(7)
b, e, c, a, f, d, g	(5)	(4)	(6)	(2)	(5)	(7)

Al-Kamal Notes

